Review addendum: PR #1186

Date: 2026-10-29. Identify Klein-four Galois groups and Kummer generators for relative quartics.

Original author head: 842c3896938ce249dc21494edbf37965a767bd51. The original study and every frozen source, protocol, result, failed attempt and timing cap are preserved. This addendum takes precedence over unqualified experimental-status language in the archived report.

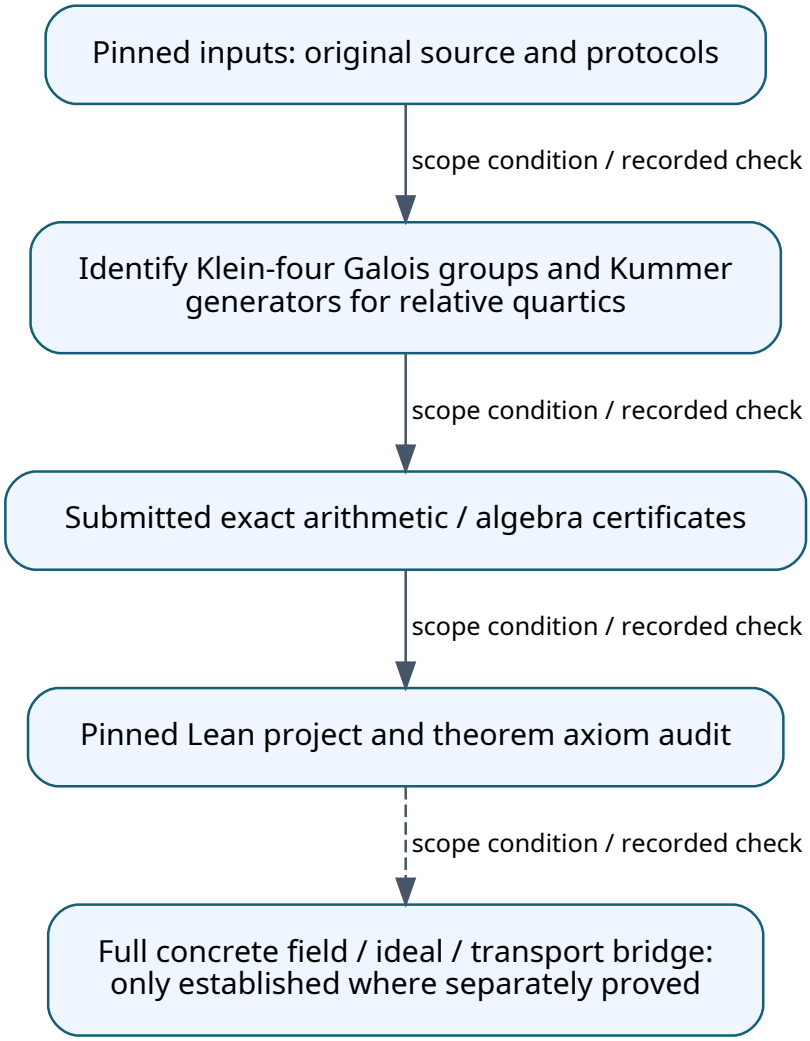
Recorded finding and its scope

The two monic irreducible quartic factors over L = K(sqrt(2)), where K is the frozen degree-18 field, each define a degree-four **Klein-four Galois extension of L**. Their fields have absolute degree 144. For each quartic, three exact cubic-resolvent roots lie in L; its polynomial discriminant has an exact square root in L from the preceding certificate. The three corresponding radicands have independent square classes of rank two, giving an explicit presentation

The paragraph above describes the retained certificate record, not a new experiment. Source: experiments/hecke-432-j1728-relative-quartic-galois-resolvent-20261026/RESULT.md (../../experiments/hecke-432-j1728-relative-quartic-galois-resolvent-20261026/RESULT.md). This review treats publication of protocol and results together as retrospective/exploratory unless an independent earlier publication witness is supplied. An archived PASS_PROSPECTIVE status is a literal historical label, not evidence of pre-observation registration. In particular, PR #1164's six-row holdout and PR #1190's 62/62 correctness checks retain their numerical results with this qualification.

Evidence-linked explanation

Evidence and proof boundaries for this study. Symbols C, B, E and K denote the frozen coordinate models or coefficient field, not newly registered subgroup instances. This diagram records algebra/certificate dependencies; it does not introduce a verified isogeny edge, EC1 identity, ISO1 route, or ECDLP transport claim.



Editable source: 1186.dot (1186.dot). Report and visual: 1186.pdf (1186.pdf).

Replay and proof map

The exact manifest binds 19 files in this study. The shared project pins Lean leanprover/lean4:v4.35.0-rc4 and Mathlib 021c46bdd042f67b538b9f9871d556a93713a9a9. All 37 original modules, 113 requested declaration reports and 142 noninternal theorems passed the integrated proof replay; this does not formalize every concrete Sage/PARI field or ideal interpretation. The source manifest and proof map are tools/ci/quartic_certificates.json (../../tools/ci/quartic_certificates.json) and the original integration report (../ci-repair-quartic-20261027.md).

This study's original modules: experiments/hecke-432-j1728-relative-quartic-galois-resolvent-20261026/formal/CubicResolvent.lean.

A new portable replay ran all 42 original integer auditors successfully in a disposable canonical-layout container. The checkout was read-only and all 1,404 original input hashes were verified before and after replay. See portable replay evidence (portable-replay/results.json). The separate field-index successor runner uses new output directories, unchanged source/protocol bytes, original timeout limits and the checked Sage launcher. Preparation and I/O controls are not a new Sage experiment.

Remaining obligations and graph review

Arithmetic table certificates and universal ring/permutation identities retain their precise hypotheses. Concrete number-field encodings, integral bases, ideals, Hensel lifting and Frobenius interpretation require their own bridges. No security, speedup, prospective-validation or general impossibility claim is added.

- Open, indexed obligation: hecke432-galois-partition-kummer-bridge (../../problems/hecke432-galois-partition-kummer-bridge.md).

The candidate catalog, volcano-IC figure set and paper figures were checked for affected links. This is a review and portability correction with no new surviving graphable mathematical fact; canonical graphs remain unchanged. The precise checked-path inventory is graph-review.json (graph-review.json).